

Ex.no:8	MINI PROJECT – TOURISM : URBAN TRAFFIC DATA ANALYTICS

Domain and Problem Statement:

Domain: Tourism - Urban Traffic & Mobility Analytics

Problem Statement: Urban transportation systems face challenges due to fluctuating traffic volume, varying vehicle speeds, changing traffic density, weather conditions, and accident occurrences. This project addresses these challenges by transforming raw traffic records into structured business intelligence using Power BI and Tableau to evaluate traffic trends, location-wise vehicle volume, traffic density, vehicle types, accident patterns, weather impact, and traffic risk levels.

Dataset Description:

Dataset Name: Urban Traffic Dataset (TrafficData).

Granularity: Each row represents a single traffic observation recorded for a particular location and time.

Timeframe Covered: 2025.

Key Entities & Attributes:

- **Traffic Identification:** Record_ID, Date, Time, Location
- **Vehicle Information:** Vehicle_Type, Vehicle_Count, Average_Speed
- **Traffic Conditions:** Traffic_Density, Weather
- **Accident Information:** Accident_Occurred, Accident_Severity

Data Source:

Primary Source: SQL Server database containing the TrafficData table.

Storage Infrastructure: Traffic records are stored in a local SQL Server 2022 database named UrbanPulseDB. The same SQL Server dataset is connected independently to Power BI and Tableau for analysis.

Architecture:



Program:**1. SQL Server – Data Retrieval & Preparation****SQL Server Command Prompt:**

```
docker ps  
docker start sqlserver
```

SQL Server:

```
CREATE DATABASE UrbanPulseDB;  
GO
```

```
USE UrbanPulseDB;  
GO
```

Retrieve Traffic Data:

```
SELECT *  
FROM TrafficData;
```

Verify Data:

```
SELECT COUNT(*) AS Total_Records  
FROM TrafficData;
```

Location-wise Traffic Analysis:

```
SELECT  
    Location,  
    SUM(Vehicle_Count) AS Total_Vehicles  
FROM TrafficData  
GROUP BY Location  
ORDER BY Total_Vehicles DESC;
```

Accident Analysis:

```
SELECT  
    Accident_Occurred,  
    COUNT(*) AS Accident_Records  
FROM TrafficData  
GROUP BY Accident_Occurred;
```

2. SQL Server – Database and Data Loading**SQL Server:**

```
USE UrbanPulseDB;  
GO
```

```
SELECT *  
FROM TrafficData;
```

Database Details:

```
Server: localhost,1433  
Database: UrbanPulseDB  
Table: TrafficData
```

Power BI Connection:

Get Data → SQL Server
Server: localhost\SQLEXPRESS
Database: UrbanPulseDB
Authentication: Microsoft
Table: TrafficData

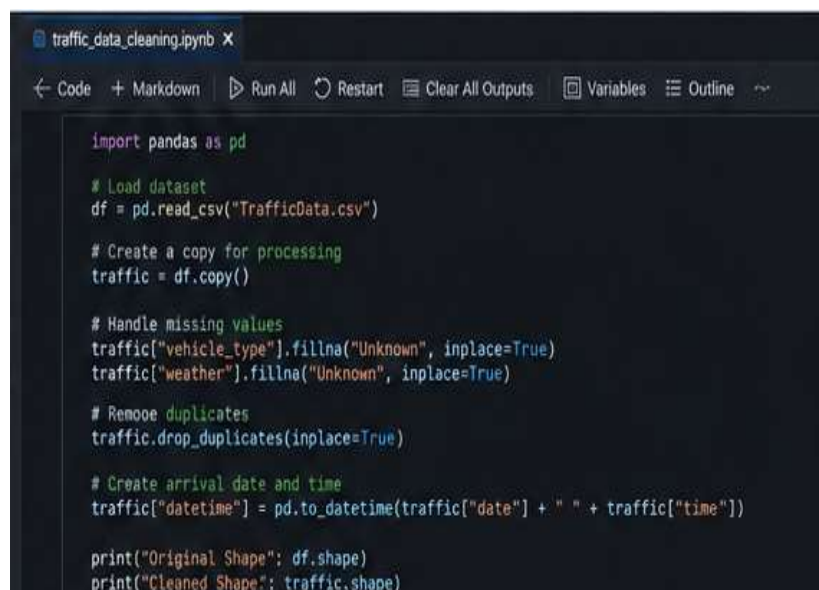
Tableau Connection:

Connect → Microsoft SQL Server
Server: localhost\SQLEXPRESS
Database: UrbanPulseDB
Table: TrafficData

Procedure:

Step 1: Data Preparation & Preprocessing (SQL Server)

- Open the SQL Server database containing the `TrafficData` table.
- Verify the traffic records and required fields.
- Check fields such as `Date`, `Time`, `Location`, `Vehicle_Type`, `Vehicle_Count`, and `Average_Speed`.
- Verify traffic condition fields including `Traffic_Density` and `Weather`.
- Verify accident-related fields including `Accident_Occurred` and `Accident_Severity`.
- Execute SQL queries to retrieve and validate the required traffic data.
- Use the validated `TrafficData` table as the common data source for Power BI and Tableau.



```
import pandas as pd

# Load dataset
df = pd.read_csv("TrafficData.csv")

# Create a copy for processing
traffic = df.copy()

# Handle missing values
traffic["vehicle_type"].fillna("Unknown", inplace=True)
traffic["weather"].fillna("Unknown", inplace=True)

# Remove duplicates
traffic.drop_duplicates(inplace=True)

# Create arrival date and time
traffic["datetime"] = pd.to_datetime(traffic["date"] + " " + traffic["time"])

print("Original Shape": df.shape)
print("Cleaned Shape": traffic.shape)
```

Figure 1: Traffic Data Preprocessing Using Python

traffic.head()

	record_id	date	time	location	vehicle_type	vehicle_count	average_speed	traffic_density	weather	accident_occurred	accident_severity
0	1	2025-01-01	08:00:00	Adyar	Car	45	42	Medium	Clear	No	None
1	2	2025-01-01	08:05:00	T.Nagar	Bus	38	35	High	Cloudy	No	None
2	3	2025-01-01	08:10:00	Velachery	Bike	60	28	High	Rain	Yes	Minor
3	4	2025-01-01	08:15:00	Anna Nagar	Truck	22	50	Low	Clear	No	None
4	5	2025-01-01	08:20:00	Guindy	Car	55	40	Medium	Cloudy	No	None

Figure 2: Traffic Dataset Preview

Step 2: SQL Database Connection & Data Ingestion

1. Open Power BI Desktop → Click Get Data → Select SQL Server.
2. Enter the Server as localhost\SQLEXPRESS and Database Name as UrbanPulseDB.
3. Select Database Authentication and enter the SQL Server credentials.
4. Select the TrafficData table and click Load to import the data into Power BI.
5. Open Tableau Desktop → Select Microsoft SQL Server.
6. Enter Server as localhost\SQLEXPRESS and Database as UrbanPulseDB.
7. Enter the SQL Server credentials and select the TrafficData table.
8. Use the retrieved SQL Server data to create dashboards and interactive visualizations in both Power BI and Tableau.

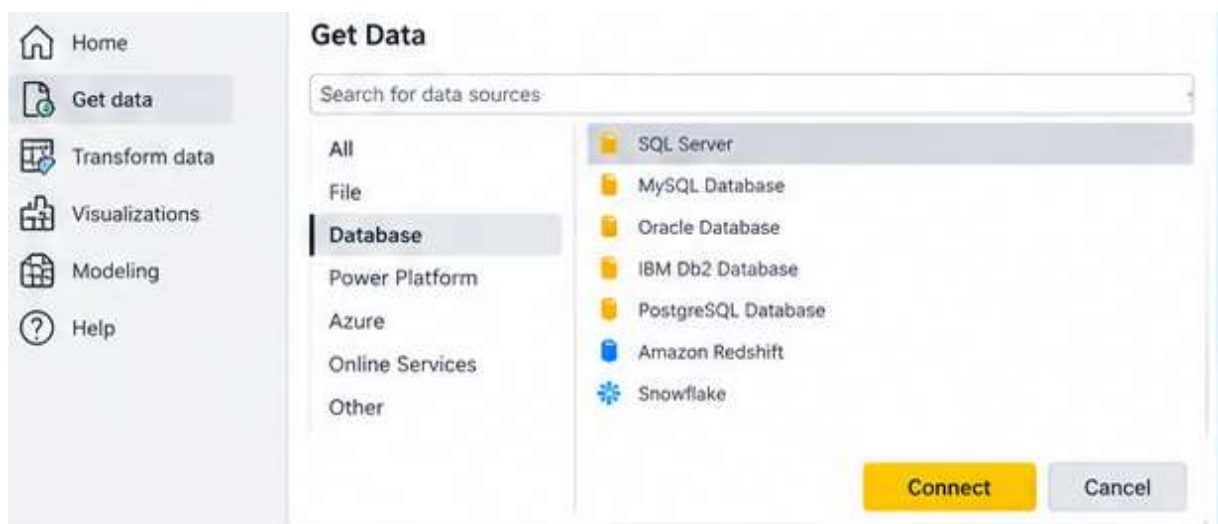


Figure 3: Power BI Data Connection

The image shows the 'Connect' dialog in Power BI. On the left, under 'To a Server', 'Microsoft SQL Server' is highlighted. The main panel is titled 'Microsoft SQL Server' and contains the following fields:

- Server: localhost,1433
- Database: UrbanPulseDB
- Authentication: SQL Server Authentication (dropdown)
- Username: sa
- Password: (masked with dots)

A 'Sign In' button is located at the bottom right of the main panel. Below the button is a link that says 'Need help connecting?'.

Figure 4: SQL Server Database Connection

Step 3: Star Schema Data Modeling (Power BI)

1. Table Splitting in Power Query:

- Duplicate the imported flat table (`TrafficData`) to create separate entities: `Fact_Traffic`, `Dim_Location`, `Dim_Vehicle`, `Dim_Date`, and `Dim_Weather`.
- In `Dim_Location`, keep the `Location` field. Remove duplicate locations and add an Index Column as `LocationID`.
- In `Dim_Vehicle`, keep the `Vehicle_Type` field. Remove duplicates and add an Index Column as `VehicleTypeID`.
- In `Dim_Date`, keep the `Date` field and create date-related attributes such as `Year`, `Month`, and `Month Name`.
- In `Dim_Weather`, keep the `Weather` field. Remove duplicates and add an Index Column as `WeatherID`.
- In `Fact_Traffic`, retain the measurable traffic fields such as `Vehicle_Count`, `Average_Speed`, `Traffic_Density`, `Accident_Occurred`, and `Accident_Severity`. Map the corresponding dimension IDs and remove unnecessary duplicate text fields.
- Click Close & Apply.

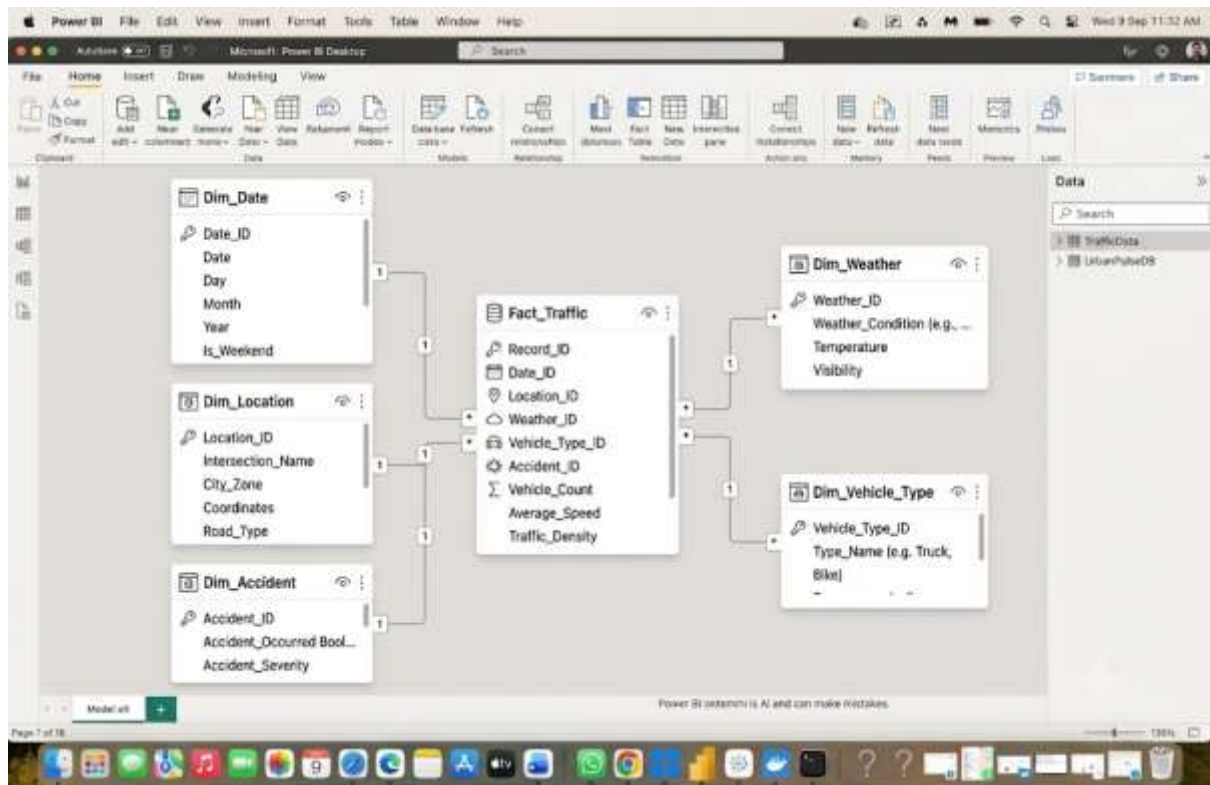


Figure 5: Power BI Star Schema Data Model

2. Establishing Relationships in Model View:

- Open the Model View tab in Power BI.
- Connect Dim_Location[LocationID] → Fact_Traffic[LocationID].
- Connect Dim_Vehicle[VehicleTypeID] → Fact_Traffic[VehicleTypeID].
- Connect Dim_Date[Date] → Fact_Traffic[Date].
- Connect Dim_Weather[WeatherID] → Fact_Traffic[WeatherID].
- Set the relationships as One-to-Many (1:*), with the dimension tables on the one side and Fact_Traffic on the many side.
- Hide foreign-key columns inside Fact_Traffic to keep the report fields pane organized.

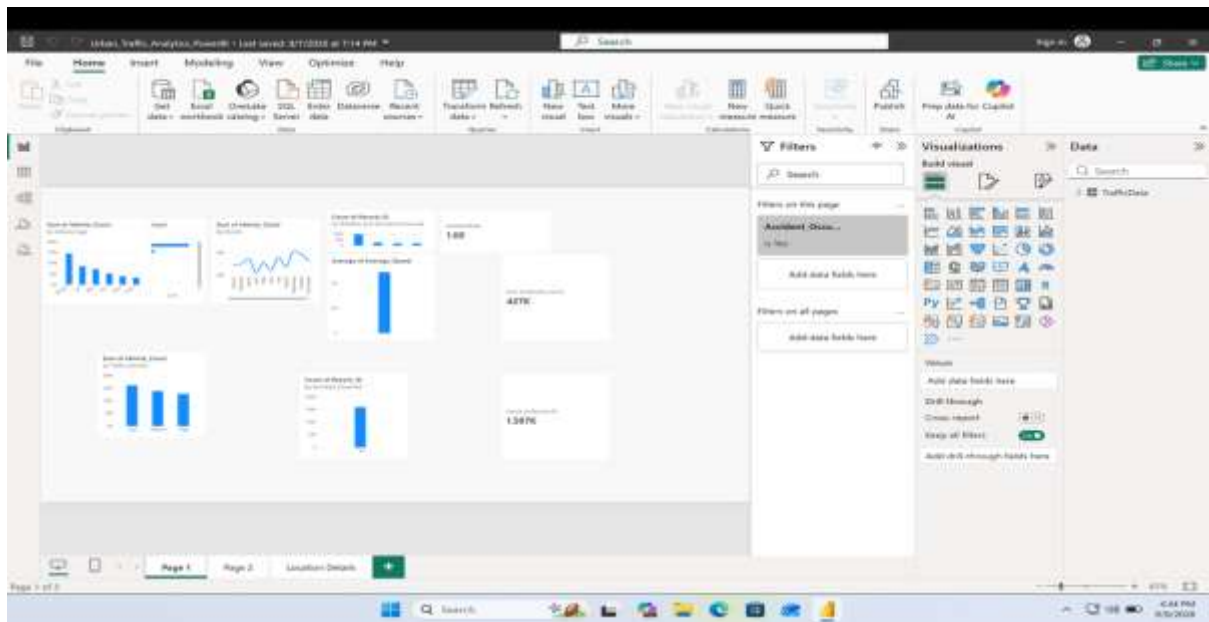


Figure 6: Power BI Traffic Analytics Dashboard

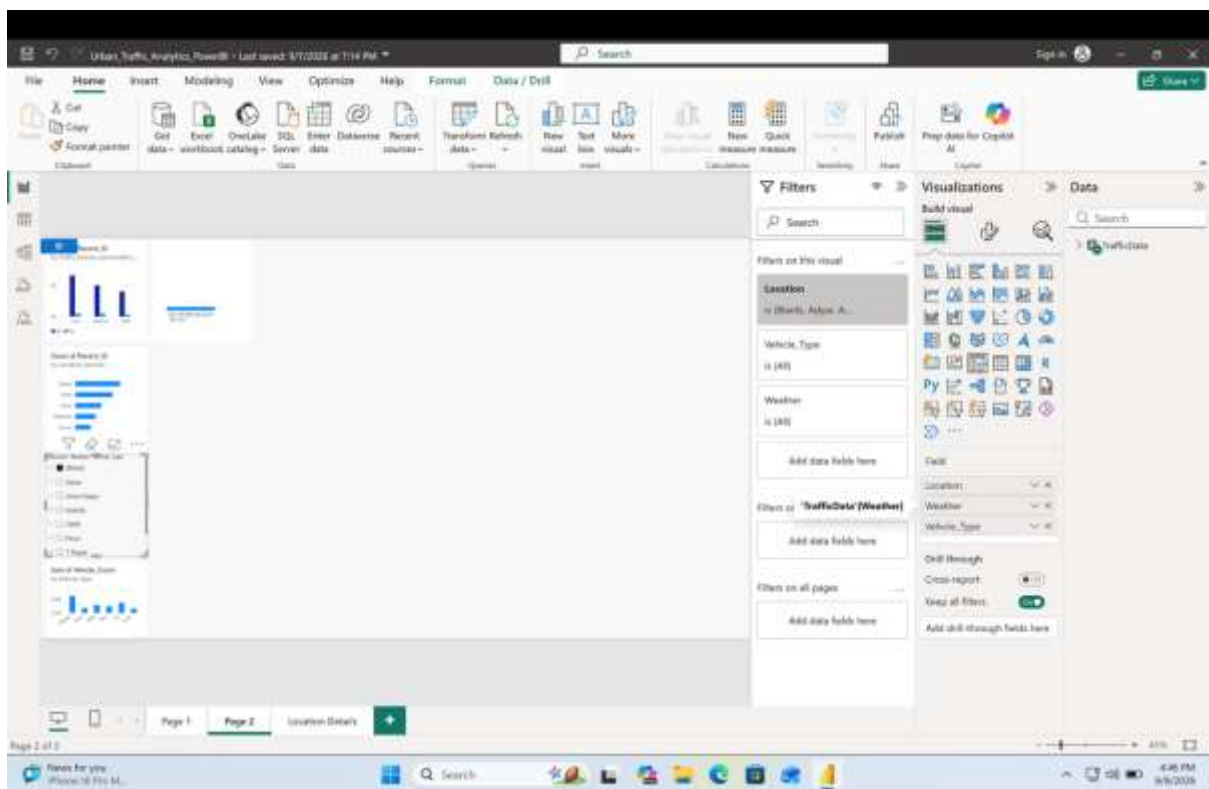


Figure 7: Power BI Traffic Analysis Visualization

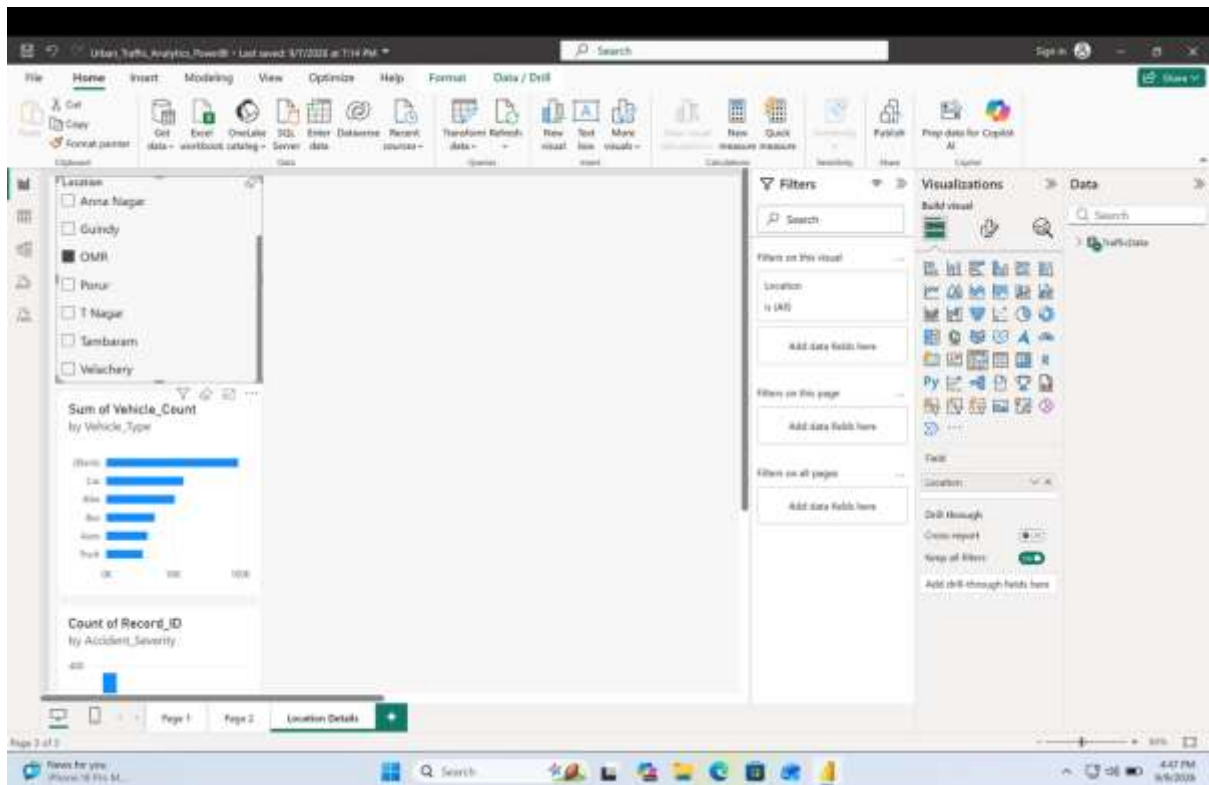


Figure 8: Location-wise Vehicle Count Analysis

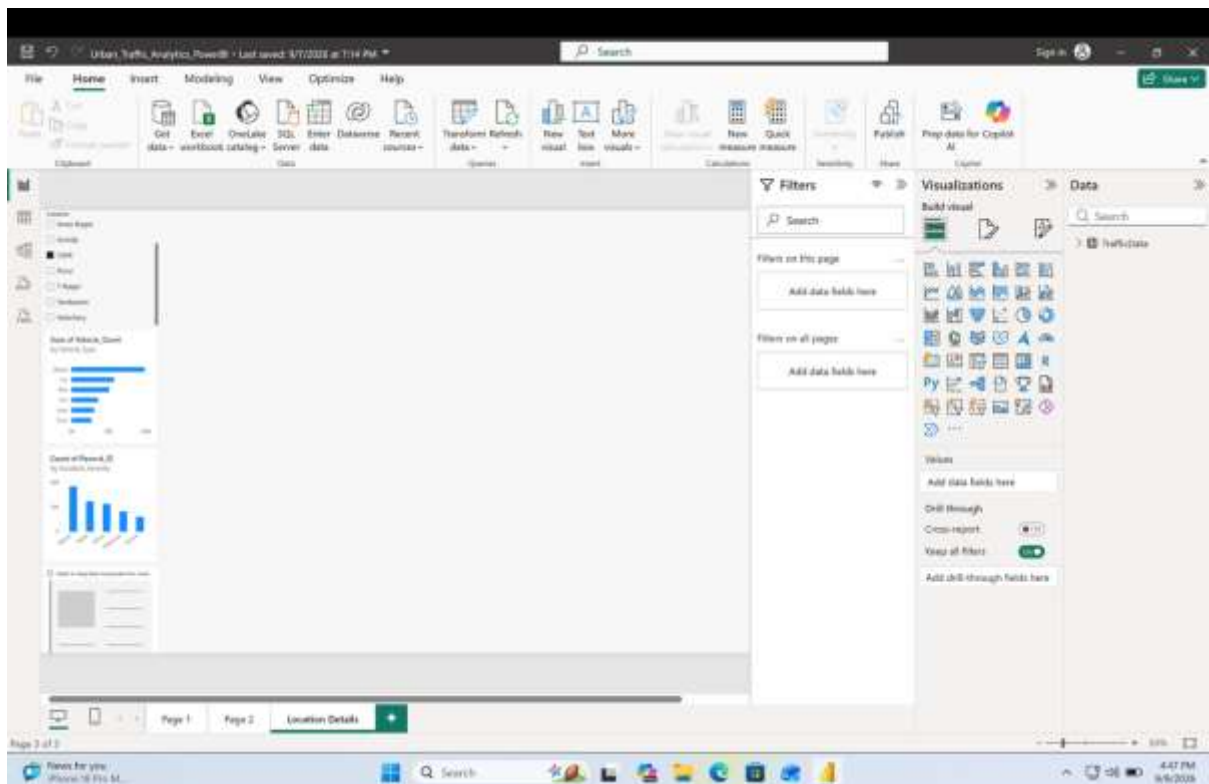


Figure 9: Power BI Dashboard with Traffic Metrics

Step 4: Core Calculations & Measures (DAX)

Power BI Measures (DAX):

The following DAX measures were created to perform key calculations for **Urban traffic pattern analysis** and to support the dashboard KPIs and visualizations.

1. Total Vehicle Count

```
Total Vehicles = SUM(TrafficData[Vehicle_Count])
```

Calculates the total number of vehicles recorded in the dataset.

2. Average Speed

```
Average Speed = AVERAGE(TrafficData[Average_Speed])
```

Calculates the average speed of vehicles across the selected data.

3. Total Accidents

```
Total Accidents =  
CALCULATE(  
    COUNTROWS(TrafficData),  
    TrafficData[Accident_Occurred] = "Yes"  
)
```

Calculates the total number of traffic records where an accident occurred.

4. Accident Rate

```
Accident Rate =  
DIVIDE(  
    CALCULATE(  
        COUNTROWS(TrafficData),  
        TrafficData[Accident_Occurred] = "Yes"  
    ),  
    COUNTROWS(TrafficData),  
    0  
)
```

Calculates the proportion of traffic records associated with accidents.

5. Average Vehicle Count

```
Average Vehicle Count =  
AVERAGE(TrafficData[Vehicle_Count])
```

Determines the average number of vehicles recorded per observation.

6. High Density Records

```

High Density Records =
CALCULATE(
    COUNTROWS(TrafficData),
    TrafficData[Traffic_Density] = "High"
)

```

Step 5: Power BI Final Executive Dashboard & Drill-Through

1. Executive Dashboard:

- Place Card Visuals across the top: Total Vehicles, Average Speed, Total Accidents, and Accident Rate.
- Add a Line Chart for the Monthly Traffic Trend using `Date` on the X-axis and `Vehicle_Count` on the Y-axis, with date hierarchy drill-down enabled (Year → Quarter → Month).
- Add a Clustered Bar Chart **for** Traffic Volume by Location using `Location` on the axis and `Vehicle_Count` as the value.
- Add a Clustered Column Chart for Traffic Density Analysis using `Traffic_Density` on the axis and `Vehicle_Count` as the value.
- Add a Clustered Column Chart for Accident Analysis using `Accident_Occurred` on the axis and Count of `Record_ID` as the value.
- Add a Clustered Column Chart for Traffic by Vehicle Type using `Vehicle_Type` on the axis and `Vehicle_Count` as the value.
- Add a Clustered Column Chart for Weather vs Accident Occurrence using `Weather` on the X-axis, Count of `Record_ID` as the value, and `Accident_Occurred` as the legend.
- Add **Slicers** for `Location`, `Weather`, and `Vehicle_Type` to enable interactive filtering of the dashboard.

Step 6: Tableau Data Connection & Preparation

- Open Tableau Desktop.
- Under To a Server, **select** Microsoft SQL Server.
- Enter the SQL Server connection details:
 - **Server:** localhost\SQLEXPRESS
 - **Database:** UrbanPulseDB
 - **Username:** sa
 - Enter the SQL Server password.
 - Select the TrafficData table to load the dataset onto the Tableau canvas.
 - Go to any worksheet and build the following **calculated fields**:
 - Go to any sheet and build calculated fields:
 - Total Vehicle Count

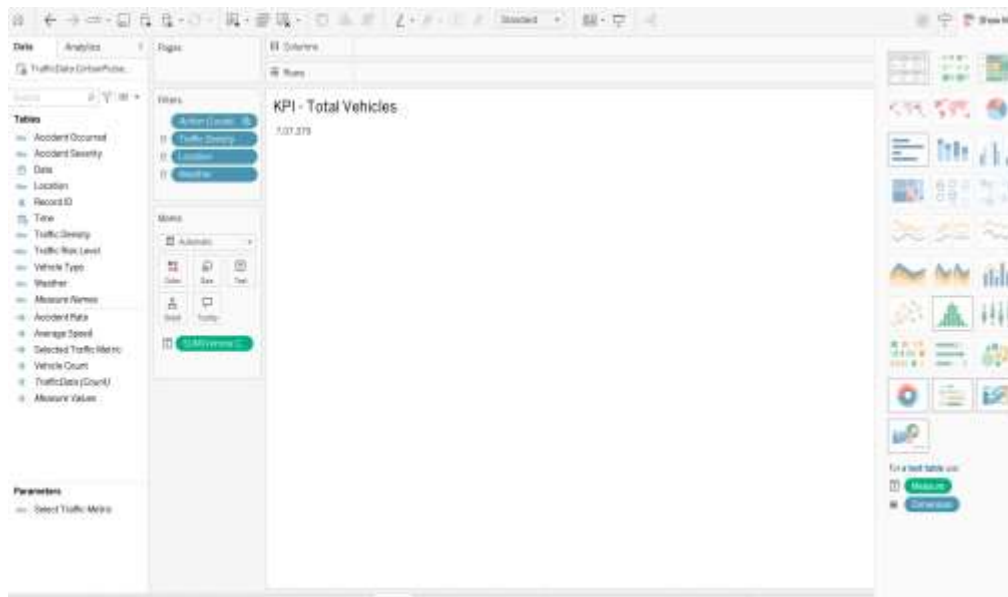


Figure 10: Tableau Total Vehicle Count

○ Average Speed

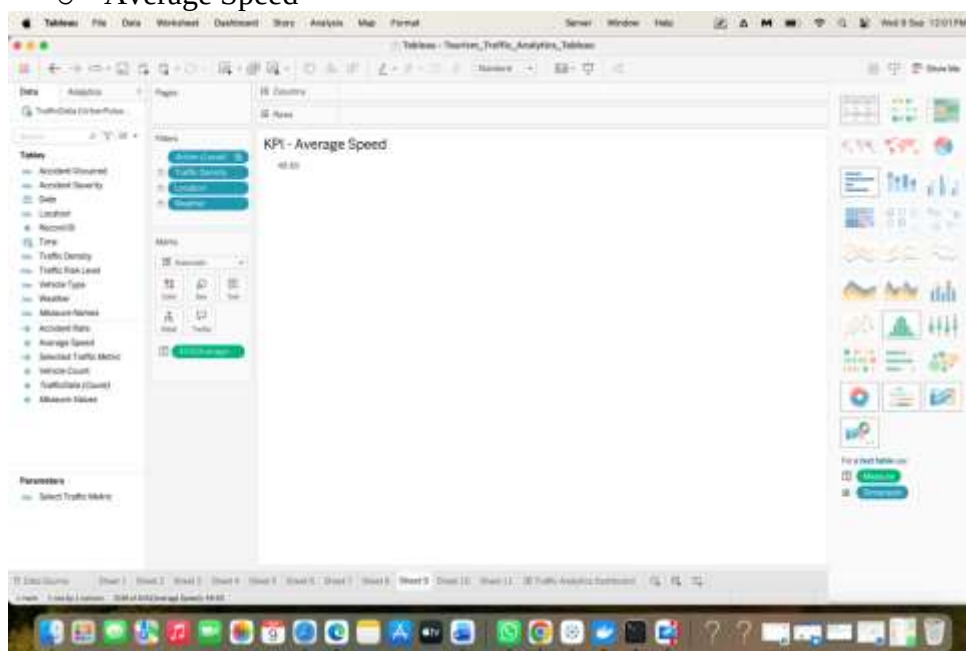


Figure 11: Tableau Average Speed

- Total Accidents

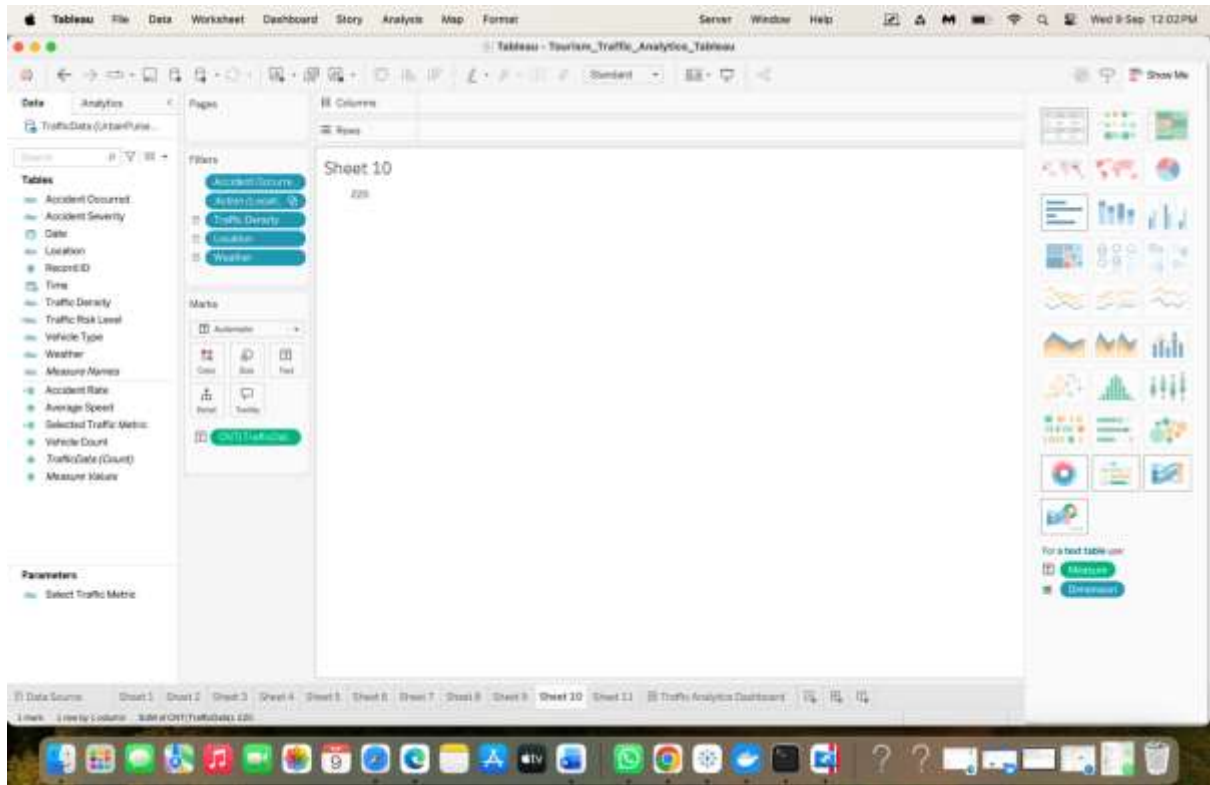


Figure 12: Tableau Total Accidents

- Accident Rate

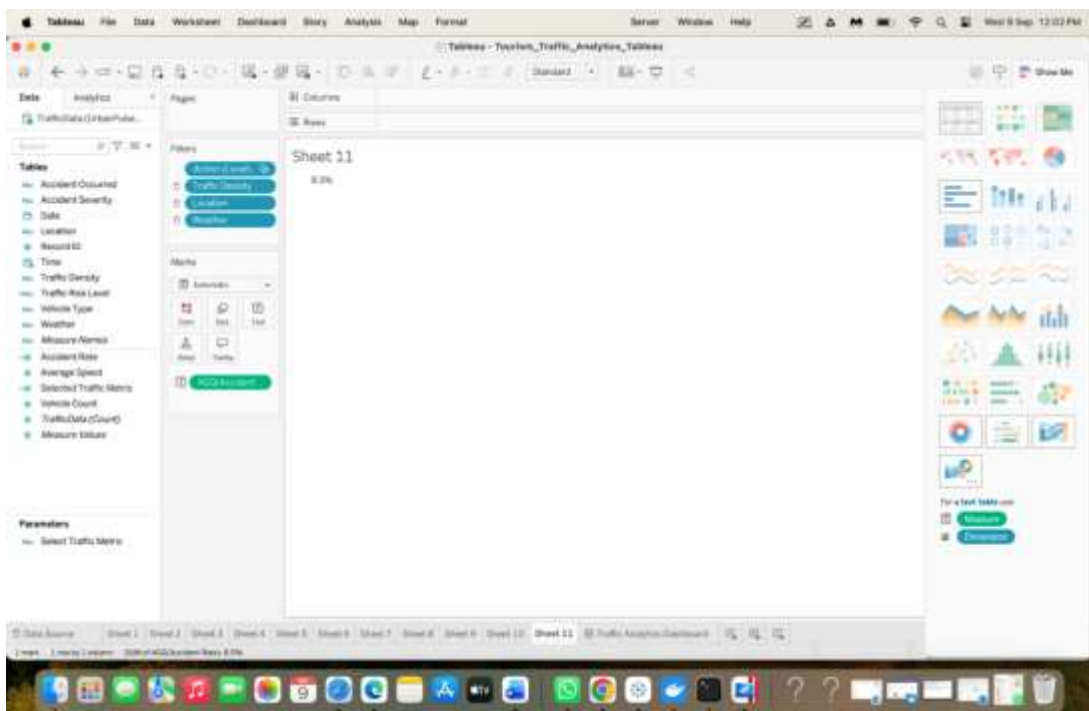


Figure 13: Tableau Accident Rate

Create a calculated field named **Dynamic Measure**:

```
CASE [Select Traffic Metric]

WHEN "Vehicle Count" THEN
    SUM([Vehicle Count])

WHEN "Average Speed" THEN
    AVG([Average Speed])

WHEN "Accident Count" THEN
    SUM(
        IF [Accident Occurred] = "Yes"
        THEN 1
        ELSE 0
        END
    )

END
```

Step 7: Tableau Companion Dashboard & Interactive Story

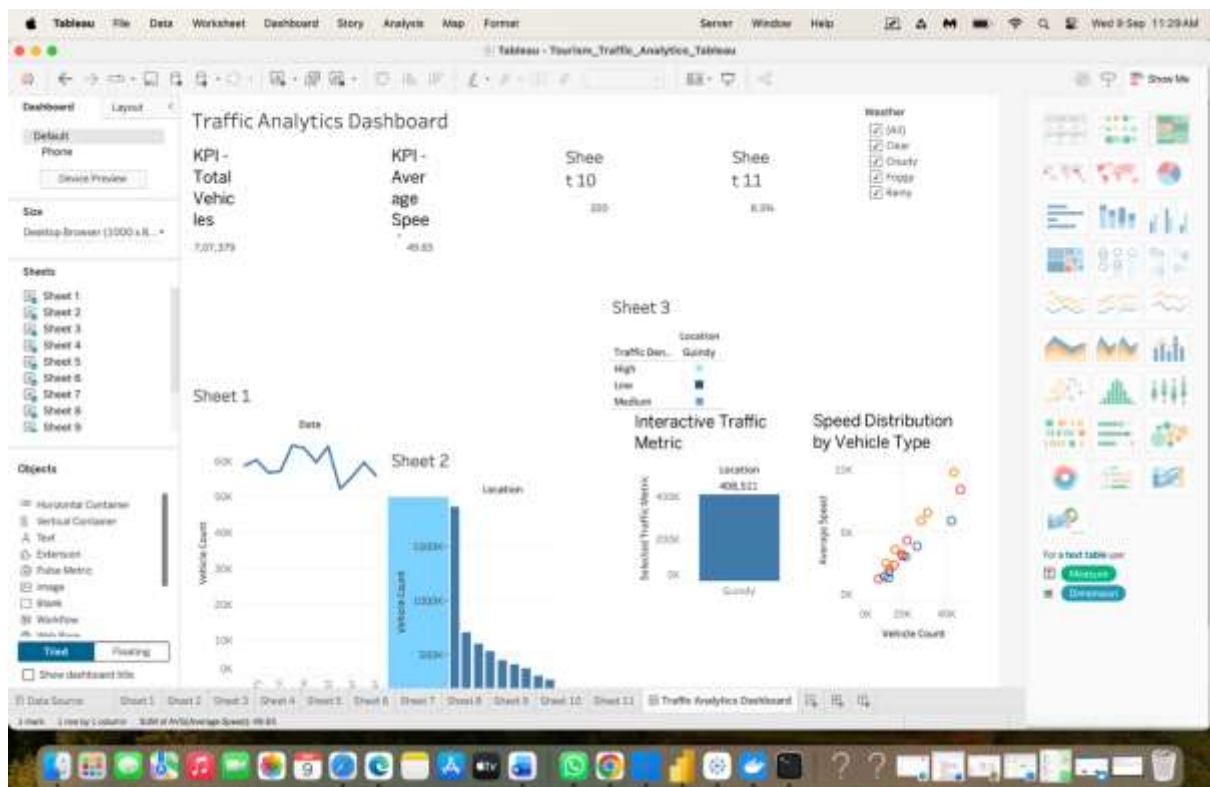


Figure 14: Tableau Interactive Traffic Analytics Dashboard

Worksheets: Build KPI Cards, Monthly Traffic Trend Line Chart (using Dynamic Measure), Traffic Density Bar Chart, Accident Severity Analysis, Traffic Risk Analysis, and **Vehicle** Count vs Average Speed Scatter Plot.

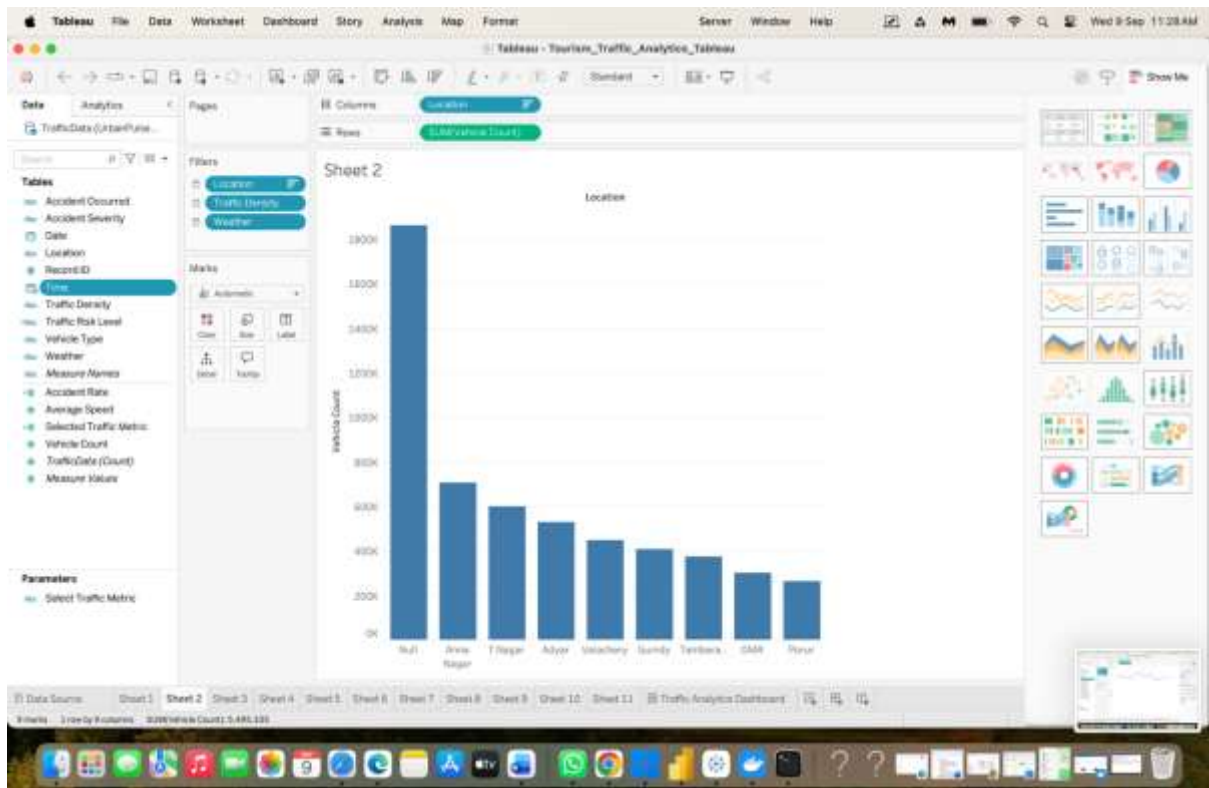


Figure 15: Location-wise Traffic Volume Analysis

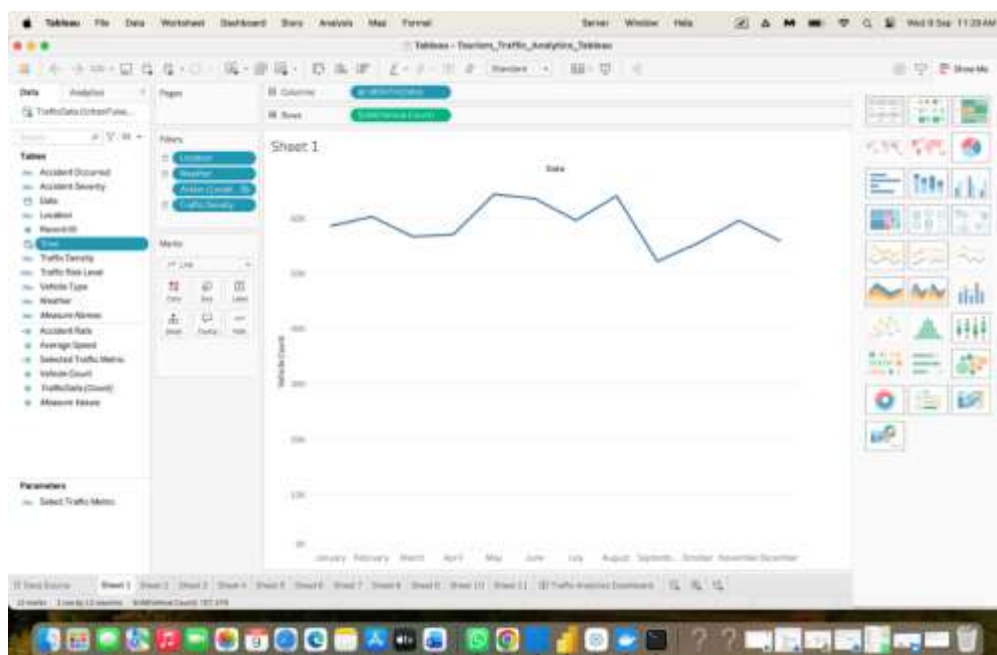


Figure 16: Monthly Traffic Volume Trend

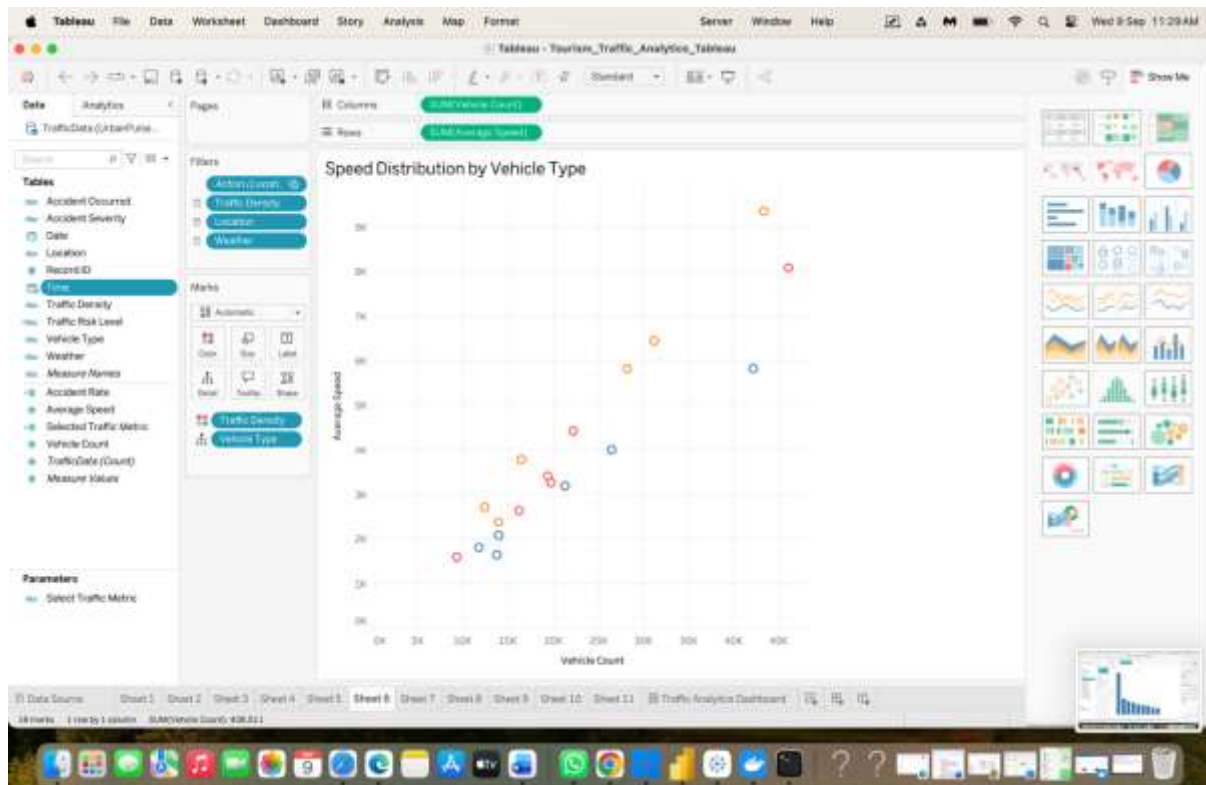


Figure 17: Speed Distribution by Vehicle Type

Observed Insights:

- **Traffic Volume & Average Speed:** The dataset records approximately **5.49 million total vehicles**, with the overall average speed providing an indication of traffic movement across the monitored locations.
- **Accident Analysis:** The dataset contains approximately **1,587 accident records**, allowing accident occurrence and severity to be analyzed across different locations and traffic conditions.
- **Traffic Density Trends:** Traffic conditions vary across **Low, Medium, and High density** levels, with high-density periods helping identify areas requiring closer monitoring and traffic management.
- **Location-wise Traffic Distribution:** Traffic volume differs across monitored locations such as **Adyar, Anna Nagar, OMR, Guindy, Porur, T. Nagar, Tambaram, and Velachery**, helping identify locations with comparatively higher traffic loads.
- **Weather & Traffic Conditions:** The dashboard enables comparison of traffic and accident occurrence across different **weather conditions**, helping identify potential relationships between weather and road incidents.
- **Vehicle Type Analysis:** Different vehicle categories contribute differently to overall traffic volume, allowing vehicle-wise traffic patterns to be compared using the Power BI and Tableau dashboards.
- **Traffic Risk Analysis:** Combining **Traffic Density** and **Accident Occurrence** helps classify traffic situations into different risk levels, supporting identification of locations and conditions that may require increased monitoring.